



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO DEPT. OF SUGAR - DHARMAPURI
DCSM LT**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

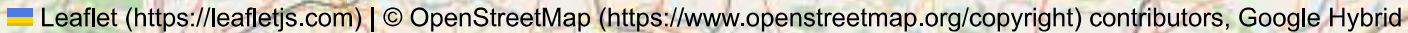
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



End: (12.322287, 78.076318)



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typically takes around 2.41 hours for trucks carrying hazardous materials.

2. Weather Conditions

- **Typical Weather:** The area generally experiences tropical weather. Summers can be extremely hot, while monsoons bring heavy rains.
- **Potential Hazards:** Monsoon season often leads to flooding and slippery roads. Visibility issues due to fog can occur during winter mornings.

3. Traffic Patterns

- **Peak Hours:** Between 7:30-10:00 AM and 5:00-8:00 PM. Significant congestion is noted in towns like Dharmapuri.
- **Congestion-Prone Areas:** Urban centers and toll booths along NH44 often experience delays.

4. Road Quality and Infrastructure

- **NH44:** Well-maintained with four-lane sections; however, occasional roadworks might impede smooth movement.
- **State Highways:** Mixed conditions. Potholes are occasional, especially post-monsoon.
- **Local Roads to Palacode:** Narrower and occasionally under-maintained, requiring cautious driving.

5. Alternative Routes

- **Emergency Detour:** Via Erode and Sivaganga for avoiding NH44; significantly longer but useful in the event of roadblocks or closures on the primary route.

6. Local Regulations

- **Hazardous Material Transport:** Restricted during peak hours in urban areas and disallowed in environments with dense residential settlements.
- **Permits:** Required for transporting specific hazardous materials across state lines.

7. Historical Incidents

- Notable accidents involve vehicle collisions and rollovers, often due to overspeeding or vehicle overloading. Minimal incidents are directly linked to hazardous materials.

8. Environmental Considerations

- **Sensitive Areas:** Pollution control zones around agricultural lands and water bodies.
- **Protected Areas:** Drivers must be mindful near forested regions closer to Dharmapuri.

9. Communication Coverage

- **Reliable Signal:** Predominantly along NH44.
- **Dead Zones:** Some patchy coverage in hilly sections and rural areas approaching Palacode.

10. Emergency Response Times

- **Urban Sections:** 15-30 minutes due to proximity to service centers.
- **Rural Areas:** Up to 45-60 minutes, especially if connectivity issues arise.

11. Overall Risk Assessment

- **Medium Risk:** Combination of effective infrastructure on NH44 with challenges posed by local road conditions and weather variability.
- **Mitigation:** Ensure compliant speed, remain updated with weather forecasts, and maintain open communication with control centers for any anomalies.

Summary

This route, though mostly reliable, requires strategic navigation around peak times and adverse weather conditions. Drivers should maintain vigilance regarding road quality issues, primarily on secondary and tertiary roads. Emergency preparedness is critical, with routine checks on vehicle status and communication devices to mitigate any potential risks en route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.64180, 78.11808	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.64325, 78.11993	30 KM/Hr
14	Turn	Medium	11.69519, 78.10775	30 KM/Hr
15	Turn	Medium	11.73958, 78.05105	30 KM/Hr
16	Turn	Medium	11.75835, 78.05016	30 KM/Hr
17	Turn	Medium	11.86776, 78.08342	30 KM/Hr
18	Turn	Medium	11.87007, 78.08384	30 KM/Hr
19	Turn	Medium	11.91345, 78.05865	30 KM/Hr
20	Turn	Medium	11.92545, 78.06496	30 KM/Hr
21	Turn	Medium	11.93691, 78.05433	30 KM/Hr
22	Turn	Medium	11.95498, 78.04689	30 KM/Hr
23	Turn	High	11.95847, 78.04989	15 KM/Hr
24	Turn	Medium	11.96062, 78.04957	30 KM/Hr
25	Turn	Medium	11.96366, 78.05142	30 KM/Hr
26	Turn	Medium	11.96427, 78.05118	30 KM/Hr
27	Turn	High	11.96526, 78.04950	15 KM/Hr
28	Turn	High	11.96637, 78.04923	15 KM/Hr
29	Turn	Medium	11.96859, 78.05102	30 KM/Hr
30	Turn	Medium	12.12027, 78.13284	30 KM/Hr
31	Turn	High	12.26533, 78.08428	15 KM/Hr
32	Turn	High	12.26550, 78.08440	15 KM/Hr
33	Turn	Medium	12.27213, 78.08087	30 KM/Hr
34	Turn	Medium	12.27348, 78.08085	30 KM/Hr
35	Turn	High	12.29595, 78.07368	15 KM/Hr
36	Turn	High	12.29651, 78.07545	15 KM/Hr
37	Turn	High	12.31195, 78.07847	15 KM/Hr
38	Turn	High	12.31360, 78.07464	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
30	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
31	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
33	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
34	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
36	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
38	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
39	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
42	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
32	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
37	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
41	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
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Speed Limit: 30 KM/Hr
Coordinates: 11.86776, 78.08342



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Speed Limit: 30 KM/Hr
Coordinates: 11.87007, 78.08384



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Speed Limit: 30 KM/Hr
Coordinates: 11.91345, 78.05865



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Speed Limit: 30 KM/Hr
Coordinates: 11.92545, 78.06496



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93691, 78.05433



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95498, 78.04689



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95847, 78.04989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96062, 78.04957



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96366, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96427, 78.05118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96526, 78.04950



Risk Type: Turn
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Speed Limit: 15 KM/Hr
Coordinates: 11.96637, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96859, 78.05102



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12027, 78.13284



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26533, 78.08428



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26550, 78.08440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27213, 78.08087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27348, 78.08085



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29595, 78.07368



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29651, 78.07545



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31195, 78.07847



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31360, 78.07464